

Wrapping Up Week 1 @ Unlox

Building a fully customized WhatsApp Group Chat Analytics Engine from scratch using core computational primitives.



Learning by **Building**

Instead of relying on ready-to-use analysis packages, the engineering directive was to break down log records down to their raw syntactic elements.

By tackling structural boundaries, processing variations, and multi-line message buffers natively, I gained deep tactical exposure into the actual mechanics of data science frameworks.

⊘ No Pandas

⊘ No Regex

⊘ No Matplotlib

⊘ No Counter

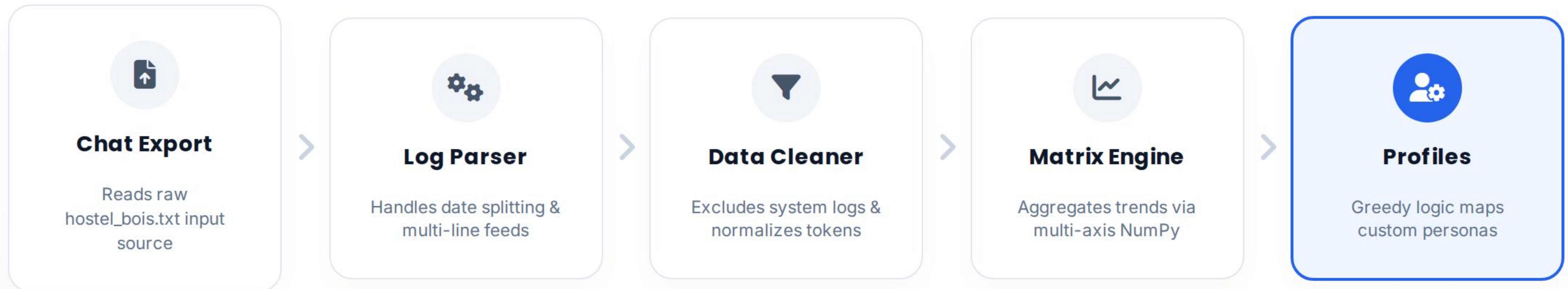


From Scratch

Building the algorithms, string indexes, and token maps entirely within pure Python & NumPy vector primitives.

GroupDNA: The Architecture

A clean, sequential processing pipeline that cleanly orchestrates raw data files into distinct, rich personality profiling models.



Engine Capabilities



Structural Log Parsing

Resolves multi-line entries, captures missing alerts, and isolating system exclusions.



NumPy Spatial Arrays

Aggregates hourly distributions within dynamic 6×24 coordinate grids.



Token Normalization

Applies localized stop-word masks and filters out non-alpha characters.



Conversational Latency

Computes inter-user response deltas while isolating long offline stretches.



Persona Assignment

Executes multi-parameter score evaluations for logical persona matches.



Plain-Text Dashboard

Generates high-fidelity visual summaries formatted for direct terminal output.

Project Highlights

3,127

Messages Processed

Isolating system alerts, deletions, and empty logs.

6 / 60

Members & Days

A closed community scope analyzing continuous interactions.

18:00

Peak Hour Volume

Max global peak registering 244 total messages.

621

Persona Top Score

Caring terms mapping 'Priya' as "THE GROUP MOM".

GroupDNA Report

ASSIGNED ARCHETYPES:

Rahul -> THE SPAMMER (4.5/burst)
Priya -> THE GROUP MOM (621.0 pts)
Aman -> THE NIGHT OWL (80.4% night)
Karan -> THE STORYTELLER (57.0 words)
Neha -> THE DRAMA QUEEN (63.3% CAPS)
Vikas -> THE GHOST (11 days silent)

✓ Done! Pure Python + NumPy Vectorization

Engineering Reflections

✓ Demystifying Core Frameworks

Building without Pandas shows how algorithms actually read strings, handle memory buffers, and count hashes behind closed abstractions.

✓ Algorithmic Shift in Thinking

Resolving multi-line parsing and conditional dates with index offsets improved my problem-solving skills much more than simply running Regex commands.

✓ NumPy Vectorization Power

Utilizing matrix functions like `.sum(axis=1)` and `.argmax()` eliminated nested tracking loops, drastically accelerating matrix executions.

"Libraries simplify development. Fundamentals build engineers."

— KEY CORE TAKEAWAY

WEEK 1 COMPLETE

On to **Week 2**

Excited to carry forward these vector habits, build more complex logic, and document my growth openly with the Unlox community.



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Follow my journey for Week 2 updates!

Image Sources



https://miro.medium.com/1*I0u9VaJUCE9kNcSqWVbsVA.png

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